

Definition of Sufficient Statistics

A Sufficient Statistic for a Parameter

- Suppose $X_1, \dots, X_n$ is a random sample from a distribution with pdf $f(x; \theta)$, $\theta \in \Omega$. Instead of listing all of the individual observations $X_1, \dots, X_n$, we might prefer to give only the statistics such as the sample mean $\bar{X}$ or the sample variance S^2 .
- Statisticians look for ways of reducing a set of data so that these data can be more easily understood without losing the meaning associated with the entire set of observations.
- Goal: Keep all information about θ .

- **Sufficient Statistic:** Consider a statistic denoted by $Y_1 = u_1(X_1, \dots, X_n)$. If the conditional probability of $X_1, \dots, X_n$ given Y_1 does not depend on θ , we call Y_1 a sufficient statistic.
- Intuitively, once we know everything about Y_1 , the distribution of another statistic, say $Y_2 = u_2(X_1, \dots, X_n)$ does not depend on θ . Hence it is impossible to use, Y_2 , given $Y_1 = y$ to make statistical inference about θ .

- Let $X_1, \dots, X_n$ denote a random sample from the distribution that has pmf

$$f(x; \theta) = \theta^x (1 - \theta)^{1-x}, \quad x = 0, 1; 0 < \theta < 1.$$

The statistic $Y_1 = X_1 + X_2 + \dots + X_n$ has the pmf

$$f_{Y_1}(y_1) = \binom{n}{y_1} \theta^{y_1} (1 - \theta)^{n-y_1}, \quad y_1 = 0, \dots, n.$$

What is the conditional probability

$$\mathbb{P}(X_1 = x_1, \dots, X_n = x_n | Y_1 = y_1)$$

- This conditional probability does NOT depend on θ .

- Let $X_1, \dots, X_n$ denote a random sample of size n from a distribution that has pdf or pmf $f(x; \theta)$, $\theta \in \Omega$. Let $Y_1 = u_1(X_1, \dots, X_n)$ be a statistic whose pdf or pmf is $f_{Y_1}(y_1; \theta)$. Then Y_1 is a **sufficient statistic** for θ iff

$$\frac{f(x_1; \theta)f(x_2; \theta) \cdots f(x_n; \theta)}{f_{Y_1}(u_1(X_1, \dots, X_n))} = H(x_1, \dots, x_n),$$

where $H(x_1, \dots, x_n)$ does not depend on $\theta \in \Omega$.

- Let $X_1, \dots, X_n$ be from a gamma distribution with $\alpha = 2$ and $\beta = \theta > 0$. Let $Y_1 = \sum_{i=1}^n X_i$. Then Y_1 has a gamma distribution with $\alpha = 2n$ and $\beta = \theta$. We have the ratio

$$\frac{f(x_1; \theta)f(x_2; \theta) \cdots f(x_n; \theta)}{f_{Y_1}(u_1(X_1, \dots, X_n))}$$

does not depend on θ .

- Let $Y_1 < Y_2 < \cdots < Y_n$ denote the order statistics of a random sample from

$$f(x; \theta) = e^{-(x-\theta)} I_{(\theta, \infty)}(x).$$

Let $Y_1 = \min\{X_i\}$ and the pdf of Y_1 is

$$f_{Y_1}(y_1; \theta) = ne^{-n(y_1-\theta)} I_{(\theta, \infty)}(y_1).$$

The ratio

$$\frac{f(x_1; \theta) f(x_2; \theta) \cdots f(x_n; \theta)}{f_{Y_1}(u_1(X_1, \dots, X_n))}$$

does not depend on θ .